TLS PROPOSAL

NOT FOR FABRICATION

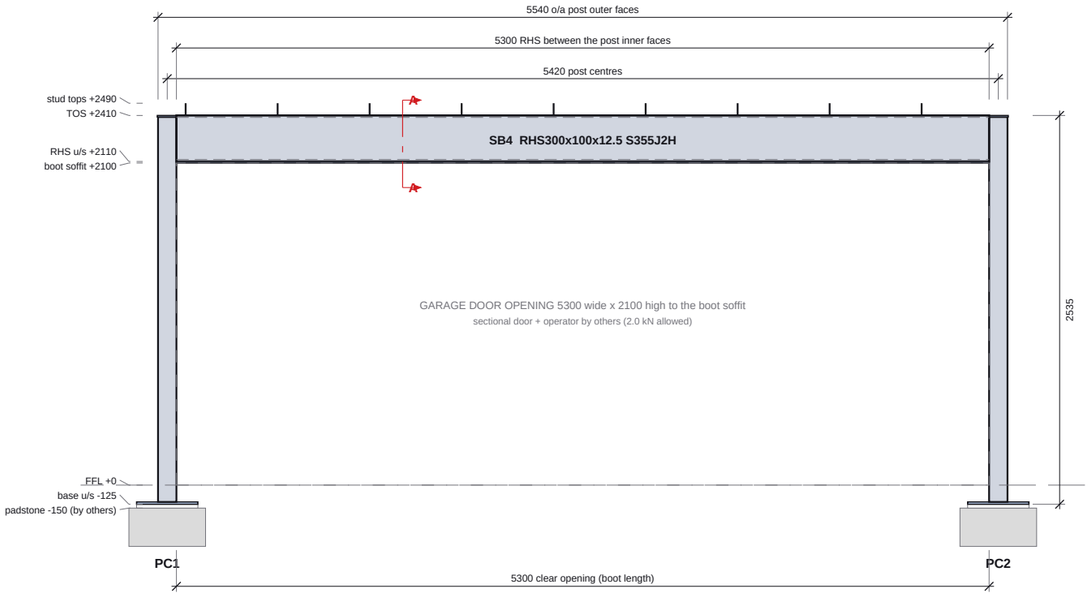
FOR ENGINEERS REVIEW

until the engineer

has reviewed + signed off

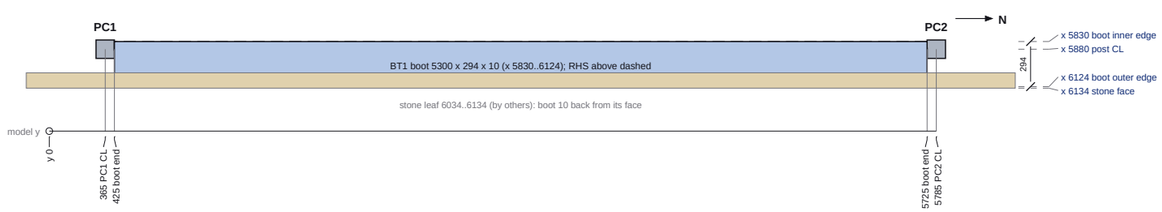
1 GP1 ELEVATION - LOOKING WEST FROM OUTSIDE

Scale 1:40 | levels above FFL; south (PC1) left, north (PC2) right



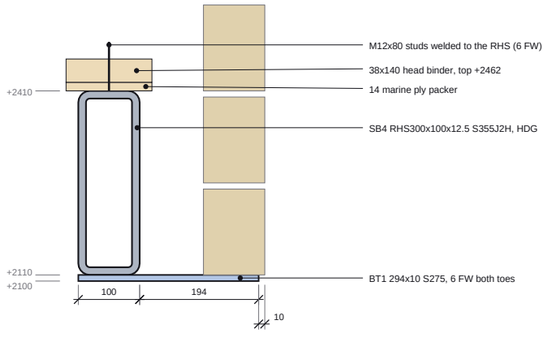
2 GP1 PLAN AT BOOT LEVEL

Scale 1:40 | outside (east) down; north right



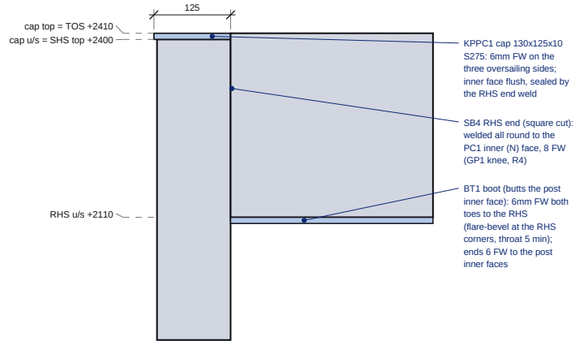
3 SECTION A-A

Scale 1:10 | looking north; inside left



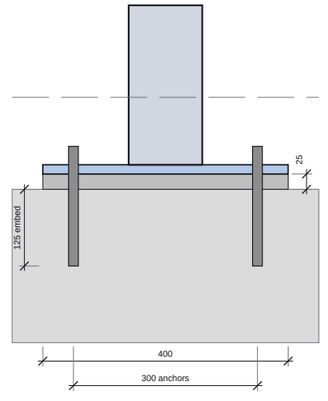
4 KNEE (PC1; PC2 HANDED)

Scale 1:10 | looking west from outside; north right



5 BASE (PC1 / PC2)

Scale 1:10 | elevation



BPPC1/BPPC2 400x200x15 S275 (200 across x 400 along the frame), 8mm FW all round
4 no. M16 8.8 HDG rod, R-KEM II, 125 embed, in Ø18 holes at x ±50 / y ±150 of the post CL (view along the frame: the two rows are in line)
25 HDG shims + non-shrink grout; padstone top -150 by others

GP1 PARTS

MARK	ITEM	LENGTH	KG
SB4	RHS300x100x12.5	5300	383.2
BT1	PL 5300x294x10 S275		122.22
PC1	SHS120x120x10	2510	84.6
BPPC1	PL 400x200x15 S275		9.30
KPPC1	PL 130x125x10 S275		1.28
PC2	SHS120x120x10	2510	84.6
BPPC2	PL 400x200x15 S275		9.30
KPPC2	PL 130x125x10 S275		1.28
GP1	**welded assembly, HDG as one piece**		**695.8**

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. GP1 R4 - PRELIMINARY, FOR ENGINEER REVIEW (TLS proposal, not the engineer's design). Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
4. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
5. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
6. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
7. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
8. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
9. FOR THE ENGINEER: loads, wind share, knee welds + post face (stiffeners in the SHS?), base fixity, padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

GARAGE PORTAL GP1 ASSEMBLY

Welded goal-post frame SB4 + PC1 + PC2 with boot BT1 (TLS proposal)

SCALE	SIZE	DATE	DRAWN
1:40 / 1:10	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-GP1	C03

Do not scale. All dimensions in mm. Check all dimensions on site.